PyTorch environment setup

PyTorch environment setup

1. System information
2. Install the dependencies
3. Install PyTorch
    3.1. Offline installation
    3.2. Online installation
4. Verify the installation

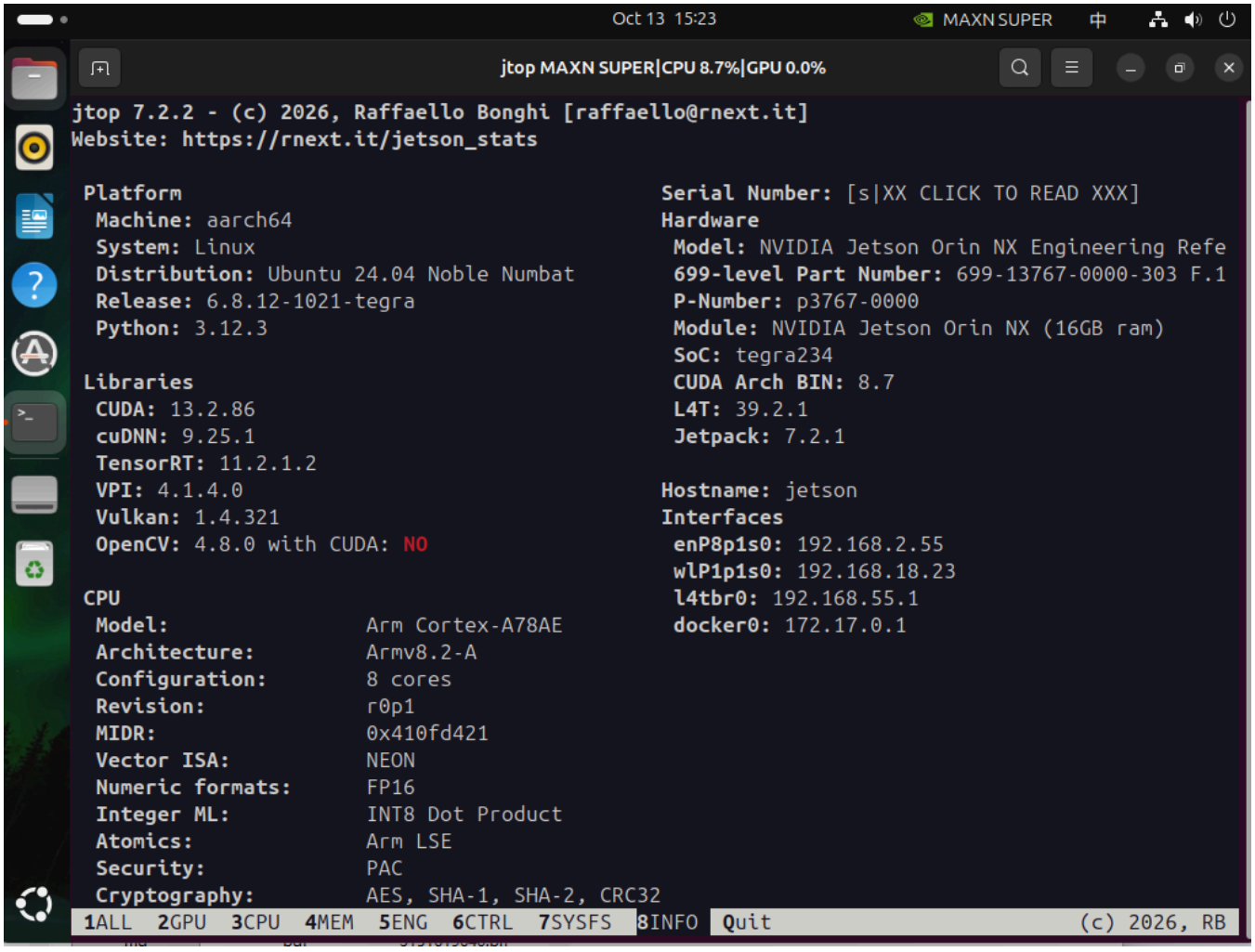
References

> **pip installation note:** Under PEP 668, Ubuntu 24.04 marks the system Python as "externally managed", so installing directly with pip may report `externally-managed-environment`. `--break-system-packages` allows pip to install packages in that environment; virtual environments usually do not need this parameter, but it is kept in this tutorial so that the commands stay consistent.

> The factory system already has the relevant environment set up, so the installation steps can be skipped.

PyTorch is an open-source deep learning framework developed by Meta (formerly Facebook). It is known for its flexibility and ease of use, and is widely used in academic research and industrial applications.

1. System information



The current system uses PyTorch `2.8.0`, Python `3.12`, ARM64, and the PyTorch CUDA build version is `12.9`.

2. Install the dependencies

```
sudo apt update
sudo apt install python3-pip libopenblas-dev -y
```

Before the installation, confirm the CUDA/cuSPARSELt dependencies according to the NVIDIA instructions for the current JetPack version; do not add the Ubuntu 22.04 CUDA software source, and do not uninstall `python3-sympy`.

3. Install PyTorch

The current system has verified that PyTorch `2.8.0` runs on CUDA `12.9`. The installation must use a wheel for NVIDIA Jetson ARM64 with CPython 3.12 that matches the current CUDA/JetPack; the PyTorch `2.5.0a0`, `cp310` or JetPack 6 wheels from the old tutorial must not be used.

Choose either the offline or the online method to install PyTorch!

3.1. Offline installation

Download the PyTorch `2.8.0` wheel that matches the current JetPack, Python ABI and CUDA version from the NVIDIA Jetson PyTorch release page.

Download URL: https://docs.nvidia.com/deeplearning/frameworks/install-pytorch-jetson-platform/index.html

```
cd Downloads/
```

```
python3 -m pip install --user --break-system-packages ./<downloaded PyTorch 2.8.0 ARM64
wheel file>
```

3.2. Online installation

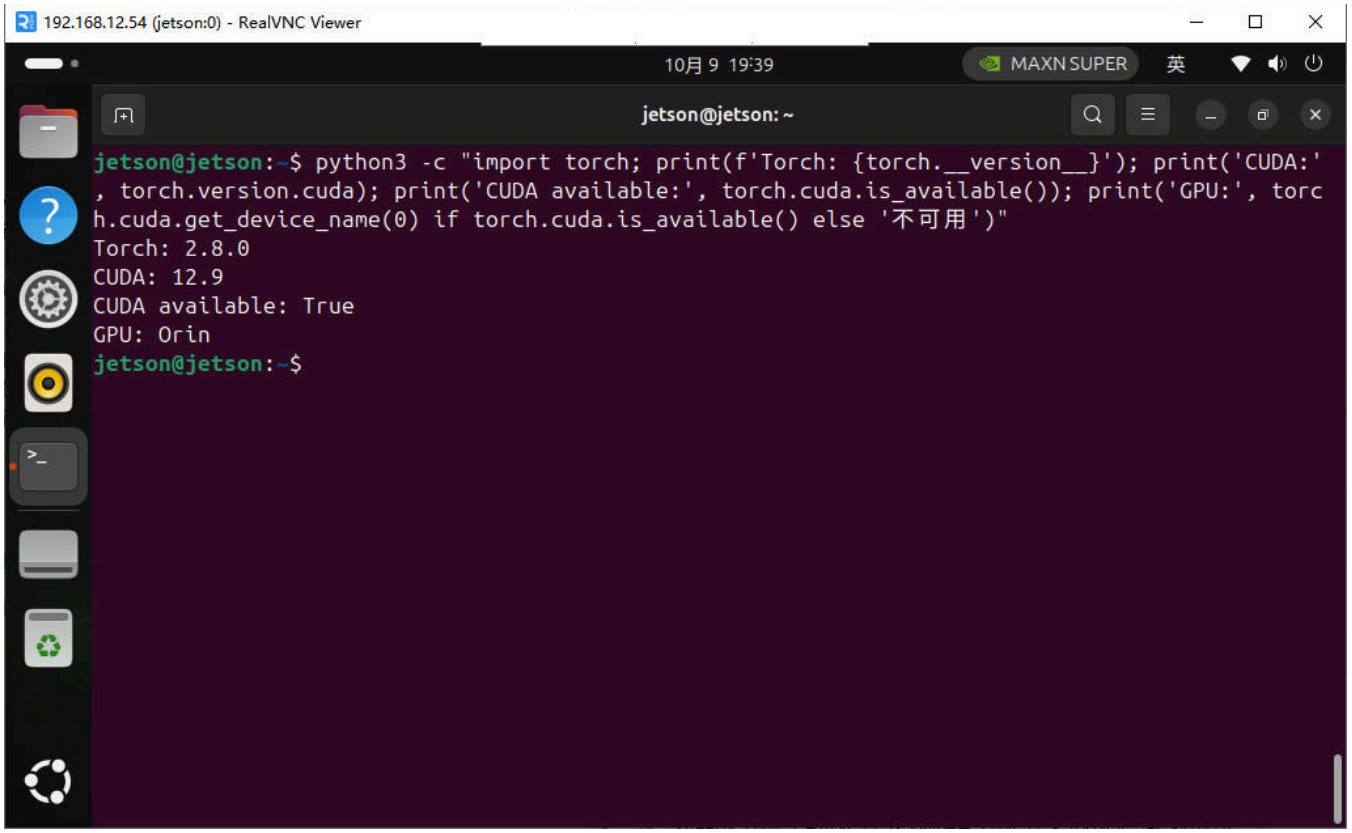
Copy the matching wheel URL from the current NVIDIA Jetson wheel release page:

```
export TORCH_INSTALL='<NVIDIA wheel URL for PyTorch 2.8.0, CPython 3.12, ARM64, CUDA 12.9>'
```

```
python3 -m pip install --user --break-system-packages --no-cache-dir "$TORCH_INSTALL"
```

4. Verify the installation

```
python3 -c "import torch; print(f'Torch: {torch.__version__}'); print('CUDA:',
torch.version.cuda); print('CUDA available:', torch.cuda.is_available()); print('GPU:',
torch.cuda.get_device_name(0) if torch.cuda.is_available() else 'unavailable')"
```



References

https://docs.nvidia.com/deeplearning/frameworks/install-pytorch-jetson-platform/index.html

https://developer.nvidia.com/cusparselt-downloads